

India's Export Trade Performance Analytics

Objective:

This project focuses on analysing India's export trade statistics using Python to identify export trends, high-performing commodities, top destination countries, and regional trade patterns. The dataset contains real-world international trade records including export values, quantities, commodity classifications, and geographic information. Through data cleaning, exploratory data analysis, and visualization techniques, the project aims to generate meaningful business and economic insights from India's global export activities.

Business Problem

India exports thousands of commodities to multiple countries and regions worldwide. However, identifying top-performing export commodities, high-value export destinations, regional trade patterns, export growth trends, quantity vs revenue relationships is difficult without proper data analysis.

Organizations and policymakers require insights into export performance to:

- Understand international market demand
- Identify profitable export sectors
- Monitor trade growth

Expected Outcome:

The project aims to:

- Analyse India's export performance over time
- Identify top export destination countries
- Determine high-performing commodities
- Study regional and sub-regional trade trends
- Compare export quantity and export value
- Generate meaningful business insights using visualization and EDA techniques

Dataset Information:

Source: [Indian Data Portal](#)

The dataset is sourced from: India Data Portal Trade/export records published through Indian government statistical systems. This is a government/public economic dataset, which makes your project look more authentic and professional compared to random practice datasets.

Dataset Size:

Metric Value

Number of Rows 532K

Number of Columns 15

Dataset Size ~ 50.6 MB

Summary of Findings:

This project analyzed India's exports to European countries using Python. The dataset contained approximately 532,000 export transactions covering multiple commodities, countries, quantities, and trade values.

The analysis showed that export trade value is highly concentrated among a small number of countries and commodities. The Kingdom of the Netherlands emerged as the most significant export destination in terms of trade value. Fuel-related products, particularly Automotive Diesel Fuel and Aviation Turbine Fuel, contributed a major share of export revenue.

Monthly and quarterly trend analysis revealed seasonal variations in export performance, with stronger export activity observed during the second quarter. Correlation analysis showed a strong positive relationship between export quantity and export value, indicating that larger shipment quantities generally generate higher trade revenue.

Overall, the export market is driven by a limited number of high-value commodities and major destination countries.

Key Insights:

1. Export value distribution is highly skewed

Most export transactions have relatively low trade values, while a small number of transactions contribute exceptionally high values. This creates a positively skewed distribution with numerous outliers.

2. Netherlands is the dominant export destination

The Kingdom of the Netherlands consistently records the highest export value among European countries, making it the most important market in the dataset.

3. Fuel products drive export revenue

Automotive Diesel Fuel and Aviation Turbine Fuel contribute significantly more export value than other commodities, indicating heavy dependence on energy-related exports.

4. Quantity and export trade values are strongly related

Correlation analysis and scatter plots show a strong positive relationship between export quantity and trade value. Higher shipment volumes generally result in higher export earnings.

5. Export performance peaks during q2 of both 2024 and 2025

Quarterly analysis shows stronger export activity during the second quarter compared with other quarters, suggesting possible seasonal demand patterns.

Types of Analysis performed:

1. Descriptive Analysis

Descriptive analysis was used to summarize and understand the dataset.

Examples:

- Dataset size and structure analysis
- Statistical summary of export values
- Distribution analysis using histograms
- Outlier identification using boxplots
- Top countries by export value
- Top commodities by export value

2. Diagnostic Analysis

Diagnostic analysis was used to understand why certain patterns exist.

Examples:

- Correlation analysis between quantity and trade value
- Commodity diversity analysis across countries
- Country-level export concentration analysis
- Commodity contribution analysis

3. Prescriptive Analysis

Prescriptive analysis focuses on actions that can improve export performance.

Possible Actions:

- Strengthen trade relationships with high-performing countries.
- Expand exports of high-value commodities.
- Reduce dependence on a small number of export destinations.
- Explore opportunities in lower-performing European markets.

Recommendations / Decision support:

- Maintain and strengthen trade relationships with the Kingdom of the Netherlands, as it contributes the highest export value.
- Reduce export concentration risk by expanding trade with other European countries such as France, Germany, Italy, and Belgium.
- Continue focusing on high-value fuel-related exports while exploring diversification into additional commodity categories.
- Use seasonal trade patterns for planning production, inventory, and logistics activities ahead of peak export periods.
- Monitor commodity diversity across countries to identify new market opportunities and reduce reliance on a limited set of products.

Conclusion:

The analysis reveals that India's exports to Europe are primarily driven by a few dominant countries and high-value commodities. Export performance shows clear seasonal patterns and a strong relationship between shipment quantity and trade value. By strengthening key trade partnerships while diversifying both markets and products, exporters can improve resilience and support long-term export growth.